TOBI TITUS OYEKANMI

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RESEARCH SUMMARY

Research-focused computer scientist with interests in machine learning systems, high-performance computing and scalable AI. Focused on efficient large-scale model training and inference, including GPU-aware optimization, numerical linear algebra, and system-level performance optimization for modern distributed AI workloads.

EDUCATION

New Mexico Highlands University, USA *Expected: Fall 2026*
Master of Science in Computer Science *CGPA: 4.00/4.00*
Department of Computer Science

New Mexico Highlands University, USA *Awarded: Spring 2025*
Master of Software Systems Design *CGPA: 3.98/4.00*
Department of Media Arts and Technology

Federal University of Technology, Akure, Nigeria *Awarded: Fall 2021*
Bachelor of Engineering in Electrical and Electronic Engineering *CGPA: 3.41/5.00*
Department of Electrical and Electronic Engineering

RESEARCH INTERESTS

Deep Learning, Machine Learning Systems, Efficient LLM Training and Inference, High-Performance Computing, Large Language Models, Agentic AI Systems, Retrieval-Augmented Generation (RAG), and Memory Compute Optimization for Scalable AI Systems

RESEARCH EXPERIENCE

Master's Thesis Research
Deep Learning-Enhanced Prediction of Execution Time, Memory Usage, and Power Consumption in Heterogeneous HPC Systems *NMHU, Computer Science Spring 2026 – Present*
- Designed and built a reproducible ML/DL pipeline predicting execution time, memory usage, and power consumption across two large-scale HPC job-trace datasets: F-DATA (Fugaku, 25.9M CPU-only jobs) and PM100 (Marconi100, 231K CPU+GPU jobs).
- Engineered a submission-time vs. post-hoc feature taxonomy (Tier A/B) to eliminate data leakage between features available before job execution and those only measurable afterward, directly shaping what the model's predictions can honestly claim.
- Developed a four-check feature-vetting methodology (missingness semantics, effect size, redundancy, user-concentration bias) to identify and exclude features whose apparent predictive power was driven by single-user submission patterns rather than generalizable signal.
- Implemented and evaluated analytical performance-modeling baselines: a Hierarchical Roofline model for F-DATA (validated FLOP/bandwidth fields against hardware specs, 99.99% agreement with the dataset's own operational-intensity classification) and a calibrated resource-utilization power model for PM100 ($R^2 = 0.91$ against a naive baseline's $R^2 = 0.10$).
- Applied empirical, data-driven dimensionality reduction (PCA on 384-dim job-name embeddings) after identifying that a proposal-stage default component count captured only 68% of variance, replacing it with a variance-threshold-driven selection method.
- Building toward a comparative evaluation of classical ML (Random Forest, XGBoost, LightGBM), deep learning (FNN, LSTM, TCN), and hybrid analytical-DL architectures against these baselines.

NSF BioPACIFIC MIP PREM Research Program *NMHU, Computer Science, Summer 2025*
- Applied high-performance computing (HPC) techniques and deep learning to design scalable and performance-efficient training and inference pipelines for modeling nanoscale material properties.

- Investigated Transformer-based representation learning for high-dimensional numerical data.
- Built preprocessing and feature engineering pipelines for experimental workflows using parallel computation and hardware-aware optimization.
- Investigated computational bottlenecks in large-scale pipelines and explored strategies to improve compute utilization and scalability on modern hardware.
- Explored trade-offs between computational cost, convergence efficiency, and model accuracy in large-scale training pipelines.
- Conducted performance and scalability analysis of training and inference pipelines, focusing on computational efficiency and resource utilization.

Master's Research Project – AI Music Analyzer *NMHU, Software Systems Design, Spring 2025*
Research Framing (see also Software Engineering Projects below for the technical implementation)
- Led research on AI-powered music analysis, focusing on pattern extraction, feature representation, and intelligent audio understanding.
- Designed a system integrating multiple AI models for structured audio analysis, including source separation and transcription.
- Explored the potential of extending audio pattern analysis toward anomaly detection and intelligent signal interpretation.
- Optimized inference and audio processing pipelines for latency, memory efficiency, and scalability under resource constraints.

Research Assistant – Machine Learning for Environmental Radiation Analysis *NMHU, Computer Science Spring 2024-2025*
- Conducted machine learning research on the prediction and classification of natural background radiation (alpha, beta, and gamma) using field-collected environmental sensor data from Abuja, Nigeria.
- Designed supervised learning pipelines in Python for numerical and time-series datasets, including data preprocessing, feature encoding, and binary classification of radiation exposure (harmful vs. harmless).
- Trained and evaluated multiple machine learning models (Random Forest, Naïve Bayes, SVM, Kernel SVM), achieving up to 94% test accuracy and 96.9% cross-validation accuracy.
- Assessed model performance using confusion matrices and standard evaluation metrics to compare algorithm effectiveness for radiation risk classification.
- Collaborated with researchers in radiation physics to align machine learning outputs with scientific interpretation and regulatory exposure thresholds.
- Co-authored a peer-reviewed publication in *ASRJETS* based on this research (see Publications).

Student Research Collaboration – ExploreAI *NMHU, Software Systems Design, Fall 2023*
Research Framing (see also Software Engineering Projects below for the technical implementation)
- Investigated retrieval-augmented generation (RAG) using large language models for contextual recommendation systems.
- Designed a semantic retrieval pipeline using vector search for aligning user queries with unstructured data.
- Evaluated system performance in terms of contextual relevance, retrieval accuracy, and response quality.
- Constructed a domain-specific knowledge base from user-generated data and integrated it into the retrieval pipeline.
- Evaluated latency and retrieval efficiency trade-offs in the RAG pipeline, identifying system bottlenecks in embedding search and response generation.
- Collaborated with faculty to assess system performance in real-world query scenarios.

Undergraduate Research Project – Smart Socket System *FUTA, Dept. of Electrical Eng., Fall 2021*
- Co-developed a smart socket for real-time energy monitoring in Nigeria's low-voltage distribution systems.
- Used current and voltage sensors with relay switching to monitor load and control power delivery.
- Handled embedded C programming, data logging, and testing of the microcontroller-based system.
- Collaborated on circuit simulation and contributed to project presentation and documentation.

SOFTWARE ENGINEERING PROJECTS

Agentic Debugger *Summer 2026*
Technologies: Python, PyTorch, Transformers, Qwen2.5-Coder-32B-AWQ (AWQ quantization), CUDA
- Designed a multi-agent framework for Automated Program Repair (Analyzer, Fixer, Sandbox Executor, Evaluator, Critique) that decomposes debugging into specialized, collaborating LLM agents rather than a single prompt-to-patch

pipeline.
- Built a 12-strategy mutation engine on top of OpenAI's HumanEval benchmark to generate 164 realistic, compile-checked bug injections spanning six exception classes (TypeError, NameError, AssertionError, IndexError, AttributeError, and wrong-output logic errors).
- Engineered a blended repair-scoring system (40% heuristic, 60% LLM-judged intent/root-cause/regression signals) with explicit override rules to resolve known LLM evaluator failure modes (e.g., misclassifying valid minimal fixes as intent violations).
- Implemented defensive, multi-stage output parsing (direct JSON extraction, regex fallback, compile-time validation) and a bounded critique-guided retry loop, eliminating truncation and syntax-failure classes that had capped early iterations at 92.7% pass rate.
- Achieved 95.1% PASS@1 (156/164) with an average repair score of 0.940 across the full benchmark, with full observability via an auto-generated interactive HTML dashboard.

Small Language Model Systems Benchmark *Summer 2026*
Technologies: Python, Hugging Face Transformers, PyTorch
- Conducted a systems-oriented benchmarking study of open-source Small Language Models (Microsoft Phi-4, Google Gemma, Qwen, TinyLlama, SmolLM) to evaluate the tradeoff between inference efficiency and model quality on consumer hardware.
- Measured throughput, latency, and memory footprint alongside qualitative task performance (reasoning, coding, summarization, instruction-following) across model architectures.
- Found that parameter count did not reliably predict deployment efficiency; smaller models achieved disproportionately higher throughput, while instruction-following quality varied independently of scale.
- Scoped follow-on work in quantized inference, CPU vs. GPU comparison, VRAM profiling, and energy-efficiency measurement, extending the study toward edge/IoT deployment feasibility

AI Music Analyzer *Spring 2025*
Technologies: Python, Flask, Librosa, Next.js, TypeScript, MongoDB, Spleeter, OpenAI Whisper, Linux
Technical Implementation (see above under Research Experience for the research framing)
- Engineered a full-stack web-based music analysis system for processing and modifying audio files.
- Built a Flask-based API for file uploads, audio feature extraction, and communication with AI models.
- Integrated Spleeter for vocal/instrumental separation and OpenAI Whisper for lyrics transcription.
- Developed a responsive frontend in Next.js & TypeScript with modular components & real-time feedback.
- Investigated system-level bottlenecks in multi-model inference pipelines, focusing on latency, memory usage, and scalability.
- Explored user-interpretable representations of audio features to support interactive analysis and improve usability for non-expert users.
- Explored GPU-aware optimization strategies and parallel processing considerations for multi-model inference pipelines.
- Designed with educational, creative, & research use cases in mind, targeting musicians and AI researchers with potential applications in anomaly detection and security-relevant signal analysis.

ExploreAI Tourist Guide *Fall 2023*
Technologies: Python, Django, LangChain, OpenAI API, React.js, Pinecone, BeautifulSoup, Tailwind CSS
Technical Implementation (see above under Research Experience for the research framing)
- Designed and implemented a retrieval-augmented generation (RAG) system using large language models for personalized recommendations via vector search.
- Scraped attractions user-generated reviews using BeautifulSoup to build a domain-specific knowledge base.
- Investigated how retrieval-augmented systems can improve transparency and user trust in AI-generated responses.
- Implemented a Django backend that integrated LangChain and Pinecone for retrieval-augmented generation.
- Used Django REST Framework to expose endpoints for frontend consumption and AI model interaction.
- Designed a semantic search pipeline to match user queries with contextual knowledge from tourist data.
- Developed a responsive frontend in React with Tailwind CSS for dynamic user interaction.
- Built the retrieval pipeline to extend cleanly to new domains, including intelligent data analysis and security-oriented applications.

AI Door Assistant *Fall 2023*

Technologies: Python, Flask, OpenCV, Twilio API, HTML/CSS

- Designed a computer vision-based smart assistant to detect motion at doors and trigger alerts.
- Used OpenCV to process real-time webcam streams and detect facial patterns or unfamiliar movement.
- Developed Flask backend for managing alert logic, video frame handling, and activity logging.
- Integrated Twilio API to send real-time SMS notifications for unauthorized access events.
- Designed the system for home security research and behavioral interaction experiments.

Shopping Assistant Using OpenAI API *Fall 2023*

Technologies: Python, OpenAI GPT API, BeautifulSoup, Pandas, Command-Line Interface (CLI)

- Built a Python-based command-line application to provide personalized product recommendations.
- Engineered a GPT-based recommendation logic that interprets user intent, price preferences, and product features.
- Parsed e-commerce data using BeautifulSoup to extract product names, prices, and descriptions.
- Used Pandas for data filtering, ranking, and formatting to generate structured suggestions.
- Designed as a lightweight prototype for future integration into web apps or browser-based shopping assistants.

TEACHING EXPERIENCE

Teaching Assistant *New Mexico Highlands University, Department of Computer Science*

- Introduction to Java Programming — Spring 2026
- Unix Operating Systems — Fall 2025
- Physics Laboratory I — Fall 2025
- Algebra Physics I — Fall 2025
- Artificial Intelligence — Spring 2025
- Software Engineering — Fall 2024

PRESENTATIONS AND PUBLICATIONS

Presentations

- *27th Thesis Presentation, Department of Software Systems Design, New Mexico Highlands University (Spring 2025)*: Presented *Development of an Interactive Music Analyzer for Musicians Using Artificial Intelligence.*
- *Deep Convolutional GAN for Image Generation (Spring 2024)*: Presented the design and implementation of Generative Adversarial Networks (GANs) for high-quality image synthesis, highlighting advancements in deep convolutional architectures.
- *44th Annual Conference of the Nigerian Institute of Physics (Spring 2023)*: Presented *Detection and Interpretation of X-ray Scans for the Presence of Pneumonia Using Convolutional Neural Network.*
- *44th Annual Conference of the Nigerian Institute of Physics (Spring 2023)*: Presented *Simulation Prediction of Background Radiations using Machine Learning.*
- *Radix Sort (Spring 2023)*: Delivered a presentation on the fundamentals and implementation of Radix Sort algorithms, including practical examples for undergraduate students.

Selected Publications

[J01] P. Adigun, **T. Oyekanmi**, A. Adeniyi: Simulation Prediction of Background Radiation Using Machine Learning. [PDF]

[J02] P. Adigun, A. Adeniyi, **T. Oyekanmi**: Detection and Interpretation of X-Ray Scans for the Presence of Pneumonia Using Convolutional Neural Network. [PDF]

[J03] **T. Oyekanmi**, P. Adigun, A. Adeniyi, N. Azeez: Deep Learning-based Diagnosis of Brain Cancer Using Convolutional Neural Networks on MRI Scans: A Comparative Study of Model Architectures and Tumor Classification Accuracy. [PDF]

[J04] **T. Oyekanmi**, P. Adigun, A. Adeniyi: Design and Evaluation of a Convolutional Neural Network Model for Automated Detection of Diabetic Retinopathy using Retinal Fundus Photographs. [PDF]

[J05] A. Adeniyi, **T. Oyekanmi**, P. Adigun, V. Kolawole: Application of Artificial Intelligence Models in Teletherapy: A Review of Efficacy and Ethical Implications. [PDF]

HONORS, AWARDS AND SCHOLARSHIPS

- Outstanding Contribution to Research and Innovation Award (2025), Nigeria Institute of Professional Engineers and Scientists (NIPES)
- Academic Excellence and Future Studies, Las Vegas Museum and Rough Rider Collection
- McCaffrey Endowment Award (Fall 2024 & Spring 2025), New Mexico Highlands University
- Graduate Research Assistantship, New Mexico Highlands University
- Dean's List, New Mexico Highlands University
- Awards of Excellence, Nigeria Institute of Physics (NIP) - Excellent Academic Transformation, Federal University of Technology, Akure

TECHNICAL SKILLS

Programming:	Python, C/C++, Java, JavaScript/TypeScript, SQL, CUDA
Performance/Systems:	Parallel Computing, GPU Computing, Performance Optimization
ML/NLP:	PyTorch, TensorFlow, Hugging Face, Scikit-learn, LangChain
LLM Systems:	Multi-Agent Orchestration, Model Quantization, Inference Benchmarking
Frameworks & Libraries:	Flask, Django, Django REST, React, Next.js, Node.js (Express, Mongoose), BeautifulSoup
Systems/Tools:	Docker, Git, Linux
Databases:	MySQL, MongoDB, Pinecone

MEMBERSHIPS AND PROFESSIONAL ORGANIZATIONS

Fellow Member, Nigeria Institute of Professional Engineers and Scientists (NIPES)
Professional Member, Institute of Electrical and Electronics Engineers (IEEE)
Fellow Member, Artificial Intelligence Management and Finance Institute (AIMFIN)
Proactive Member, Nigerian Institute of Physics, 2023 - Present
Member, National Society of Black Engineers, 2024 - Present
Student Member, Nigerian Society of Engineers, 2016 - Present

REFERENCES

References available upon request.